BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, KOHAT

INTERMEDIATE (12TH CLASS) EXAMINATION, 2021

PHYSICS

Past Paper 2021 • Paper Code: KT-XII-21-I

Retyped faithfully from the original board paper — same questions, no changes.

Section "A"

Time Allowed: 20 Minutes | **Maximum Marks:** 18 | **Paper Code:** Code B

Note: Q.1 Write the correct option i.e. A, B, C or D in the empty box provided opposite to each part. No marks will be awarded for cutting, erasing or over writing.

Qi. Capacitive reactance Xc = ...

(A) $2\pi/fc$ (B) $fc/2\pi$

(C) $2\pi fc$ (D) $1/2\pi fc$

Qii. For an atom of Hydrogen the radius of the first orbit is given by.....

(A) h/me^2 (B) $me/4h^2$

(C) $h^2/4\pi^2kme^2$ (D) h^2me^2

Qiii. For full wave rectifier, the minimum number of diodes required is.....

(A) 2 (B) 3

(C) 4 (D) 1

Qiv. Most of the electrons in the base of an NPN transistor flows.....

(A) Out of the base lead (B) Into the collector

(C) Into the emit (D) Into the Base supply

Qv. De Broglie waves are associated with.....

(A) Moving charged particles only (B) Moving neutral particles only

(C) All moving particles (D) All particles whether in motion or at rest

Qvi. In accordance with Bohr's theory the K.E of the electron is equal to.....

(A) $Ze^2/2r^2$ (B) Ze^2/r

(C) $Ze^2/2r$ (D) None of these

Qvii. The device in which induced emf is statically induced emf is .....

(A) Transformer
(B) Alternator
(C) AC generator
(D) Dynamo

Qviii. Half-life of given sample is 22 years. The sample will reduce to 50% of the original value after?

(A) 11 years
(B) 22 years
(C) 44 years
(D) None of these

Qix. The binding energy for nucleus A is 7.7 MeV and that for nucleus B is 7.8 MeV, which nucleus has the larger mass?

(A) Nucleus A
(B) Nucleus B
(C) Less than Nucleus A
(D) None of these

Qx. When air is replaced by a medium of dielectric constant K, the force of attraction between two charges separated by a distance r.....

(A) Decreases K times
(B) Remain unchanged
(C) Increases K times
(D) Increases K^2 times.

Qxi. One electron volt is equal to.....

(A) $9.11x10^{-18}$ J
(B) $13.6x10^{-12}$ J
(C) $1.60\ x10^{-19}$ J
(D) $6.3x10^{-18}$ J

Qxii. Ohm's law is valid for .....

(A) Thermistors
(B) Filament of bulb
(C) Semi conductor diode
(D) Metals

Qxiii. Wheatstone bridge is used to determine.....

(A) Current
(B) Un-known resistances
(C) Potential difference
(D) Electric power

Qxiv. One Weber is equal to.....

(A) N/C
(B) N/m A
(C) A.m/N
(D) N-m/A

Qxv. In order to convert Galvanometer into ammeter one has to connect ..... to Galvanometer

(A) Small resistance in series
(B) Small resistance in parallel
(C) High resistance in series
(D) High resistance in parallel

Qxvi. One Gauss (1G) is equal to.....

(A) 10^4 Telsa
(B) 10^{-4} Telsa
(C) 10^4 Telsa
(D) 10^{-8} Telsa

Qxvii. The unit of self-inductance is.....

(A) Henry (B) Curie

(C) Tesla (D) Volt

Qxviii. When an AC supply is connected to a pure resistance then.....

(A) Voltage and current are in phase (B) Voltage leads in current by 90°

(C) Current leads the voltage by 90° (D) Power loss in resistor is zero

Section "B"

Time Allowed: 2 Hours 40 Minutes. | **Maximum Marks:** 40

QQ.2. Attempt any TEN parts. All parts carry equal marks.

i. Why do heavier elements require more neutrons in order to maintain stability?

ii. Why we use Kirchoff's Law for circuit problems solution?

iii. What happen to the atomic number of nucleus that emits β-rays.

iv. If the peak value of sine wave is 1000V, what is the effective value?

v. Define mean, peak and RMS value of Sinusoidal Current and Sinusoidal Voltage.

vi. Explain the phenomenon of Electric Polarization.

vii. What is Super Conductor?

viii. Explain full wave Rectifier?

ix. Junction rule is based on law of conservation of charge explain.

x. Define and explain Magnetic Flux?

xi. Explain Hydrogen Emission Spectrum.

xii. Why the rest mass of photon be zero? Explain.

xiii. Voltages are always measured between two points, why?

Section "C"

Maximum Marks: 27

> Note: Note: Attempt any THREE questions. All questions carry equal marks.

QQ.3.

a. Define electric field intensity also find out electric field intensity at a point due to point charge.

b. Find electric field at a distance of 30cm from a 3μC point charge.

QQ.4.

a. State and explain Ampere's Law?

b. Define Galvanometer. How it is converted into an ammeter?

QQ.5.

a. Discuss the Wave Nature of particle?.

b. Find out the de-Broglie wave length of electron moving with a speed of 10^8m/s?

QQ.6.

a. Write down Mathematical Expression for energy of electron in quantized orbit.

b. Find the longest Wavelength of light capable of ionizing a Hydrogen atom. How much energy is needed to ionize a Hydrogen atom?